

# StructRAG Prompt Templates

## Global Decoding and Parsing Settings

| Setting                                    | Value Used in Experiments                                                                                                              |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Model for graph-only and StructRAG prompts | gpt-4-0613                                                                                                                             |
| Model for direct-image baseline            | gpt-4o-2024-05-13                                                                                                                      |
| Temperature                                | 0.2                                                                                                                                    |
| Top-p                                      | 1.0                                                                                                                                    |
| Maximum output length                      | 800 tokens                                                                                                                             |
| Completions per question                   | 3                                                                                                                                      |
| Invalid JSON handling                      | One formatting-only retry; remaining invalid responses are counted as incorrect.                                                       |
| Aggregation                                | Graph-level majority voting; if no complete graph receives a majority, retain edges appearing in at least two of three parsed outputs. |

## Input Placeholders

| Placeholder                 | Meaning                                                                                                  |
|-----------------------------|----------------------------------------------------------------------------------------------------------|
| {{QUESTION_ID}}             | Unique identifier of the evaluated question.                                                             |
| {{INPUT_GRAPH_JSON}}        | Diagram-derived graph $G=(V,E)$ , including nodes, accepted edges, directed flag, and optional metadata. |
| {{UNCERTAIN_EDGES_JSON}}    | Visual edge candidates $U$ with confidence scores and visual signals.                                    |
| {{RETRIEVED_TEMPLATE_JSON}} | Selected structural template $T^*$ retrieved from the canonical template library.                        |
| {{DELTA_EDGES_JSON}}        | Candidate missing edges $\Delta E = E(T^*) - E(G)$ .                                                     |
| {{TOPOLOGY_TYPE}}           | Topology category used for analysis, such as Star, Ring, Chain / Bus, Bridge / Mesh, or Other.           |
| {{IMAGE_INPUT}}             | Image payload used only by the direct-image multimodal baseline.                                         |

# Template 1: StructRAG Pattern-Aware Structural Validation

**Purpose:** Used by the full StructRAG method. The model receives  $G$ ,  $U$ ,  $T^*$ , and  $\Delta E$ , then decides whether candidate edges should be accepted or rejected.

SYSTEM:

You are a structural graph validation assistant for STEM education diagrams. Your task is to inspect a diagram-derived graph, uncertain visual edge candidates, and a retrieved abstract structural template. Return only valid JSON.

USER:

Question ID: {{QUESTION\_ID}}

Topology type: {{TOPOLOGY\_TYPE}}

Input graph  $G$ :

{{INPUT\_GRAPH\_JSON}}

Uncertain visual connections  $U$ :

{{UNCERTAIN\_EDGES\_JSON}}

Retrieved structural template  $T^*$ :

{{RETRIEVED\_TEMPLATE\_JSON}}

Candidate missing edges  $\Delta E$ :

{{DELTA\_EDGES\_JSON}}

Instructions:

1. Treat the retrieved template as structural evidence, not as an automatic answer.
2. Treat edges as undirected unless an arrow is explicitly present.
3. Check whether each candidate edge in  $\Delta E$  is structurally plausible.
4. Give stronger support to candidate edges that are also present in  $U$  with confidence  $\geq 0.40$ .
5. Do not invent nodes that are absent from  $G$  unless the input graph clearly contains an OCR node omission.
6. Do not add an edge solely because it appears in  $T^*$  if it contradicts the accepted graph and has no plausible structural or visual support.
7. Return the corrected graph using the exact schema below.

Required JSON schema:

```
{
  "status": "ok" or "error",
  "nodes": ["node_id"],
  "edges": [{"source", "target"}],
  "added_edges": [{"source", "target"}],
  "removed_edges": [{"source", "target"}],
  "edge_decisions": [
    {
      "edge": ["source", "target"],
      "decision": "accept" or "reject",
      "reason": "short structural justification"
    }
  ],
  "justification": "one concise explanation of the final graph"
}
```

## Template 2: GPT-4 Graph-Only Baseline

**Purpose:** Used for the graph-only baseline. The model receives the extracted graph and uncertain visual edges but no template retrieval result.

SYSTEM:

You are a graph reasoning assistant for STEM education diagrams. Return only valid JSON.

USER:

Question ID: {{QUESTION\_ID}}

Topology type: {{TOPOLOGY\_TYPE}}

Input graph G extracted from the submitted diagram:

{{INPUT\_GRAPH\_JSON}}

Uncertain visual connections U:

{{UNCERTAIN\_EDGES\_JSON}}

Instructions:

1. Determine whether the extracted graph appears structurally complete.
2. You may use the uncertain visual connections as possible evidence.
3. You do not have access to any retrieved template or canonical pattern.
4. Treat edges as undirected unless an arrow is explicitly present.
5. Return a corrected graph only when the missing or extra edge is strongly supported by G and U.

Required JSON schema:

```
{
  "status": "ok" or "error",
  "nodes": ["node_id"],
  "edges": [{"source", "target"}],
  "added_edges": [{"source", "target"}],
  "removed_edges": [{"source", "target"}],
  "justification": "one concise explanation"
}
```

## Template 3: Direct-Image Multimodal Baseline

**Purpose:** Used for the direct-image baseline. The model receives the diagram image and produces a graph without using the StructRAG retrieval library.

SYSTEM:

You are a multimodal diagram interpretation assistant for STEM education. Inspect the submitted diagram image and return only valid JSON.

USER:

Question ID: {{QUESTION\_ID}}

Topology type: {{TOPOLOGY\_TYPE}}

Image input:

{{IMAGE\_INPUT}}

Instructions:

1. Identify the nodes, components, junctions, and edges shown in the diagram.
2. Treat edges as undirected unless an arrow is explicitly present.
3. Include curved, partially occluded, or intersecting connections when they are visually supported.
4. Do not use any retrieved template or external structural library.
5. Return the graph represented by the diagram using the required schema.

Required JSON schema:

```
{
  "status": "ok" or "error",
  "nodes": ["node_id"],
  "edges": [{"source", "target"}],
  "uncertain_edges": [
    {
      "edge": ["source", "target"],
      "reason": "short visual uncertainty explanation"
    }
  ],
  "justification": "one concise explanation"
}
```

## Template 4: Formatting-Only Retry

**Purpose:** Used only when a response is not valid JSON. The retry is restricted to schema repair and must not alter graph content.

SYSTEM:

You are a JSON formatting assistant. You must not change the graph content.

USER:

The previous response did not conform to the required JSON schema. Reformat the response below into valid JSON only. Do not add or remove nodes or edges.

Required schema:

```
{
  "status": "ok" or "error",
  "nodes": ["node_id"],
  "edges": [{"source", "target"}],
  "added_edges": [{"source", "target"}],
  "removed_edges": [{"source", "target"}],
  "justification": "short explanation"
}
```

Previous response:

```
{{INVALID_RESPONSE}}
```

## Response Parsing and Aggregation Rules

Each prompt is submitted three times using the same model version and decoding parameters. Each response is parsed into a canonical graph by normalizing node identifiers, sorting node lists, sorting edge lists, and representing undirected edges in lexical order. If two or more parsed responses produce the same complete graph, that graph is selected as the final prediction. If no complete graph receives a majority, edge-level voting is applied: an edge is retained if it appears in at least two of the three parsed outputs. Responses that remain invalid after one formatting-only retry are counted as incorrect.

## Canonical Output Example

```
{
  "status": "error",
  "nodes": ["u", "v", "w", "x"],
  "edges": [{"u", "v"}, {"u", "w"}, {"u", "x"}],
  "added_edges": [{"u", "x"}],
  "removed_edges": [],
  "edge_decisions": [
    {
      "edge": ["u", "x"],
      "decision": "accept",
      "reason": "The retrieved star-like template supports this missing radial edge, and the visual parser flagged it as uncertain rather than rejected."
    }
  ],
  "justification": "The corrected graph preserves the accepted edges and adds the structurally plausible missing connection."
}
```